

Unit 5: Square Root Functions

Practice Worksheet

Name: _____ Date: _____ Period: _____

Section 1: Simplifying Radicals

Simplify each radical expression.

1. $\sqrt{98}$

3. $\sqrt{72}$

5. $\sqrt{200}$

2. $\sqrt{45}$

4. $\sqrt{128}$

6. $\sqrt{180}$

Section 2: Radical $\leftrightarrow$ Rational Exponents

Convert to rational exponent form.

1. $\sqrt{x^3}$

2. $\sqrt[3]{y^4}$

3. $\sqrt[5]{m^2}$

4. $(\sqrt[4]{n})^5$

Convert to radical form.

5. $x^{2/5}$

6. $y^{7/3}$

7. $a^{1/4}$

8. $b^{5/2}$

Section 3: Identifying Endpoints

Find the endpoint of each function.

1. $f(x) = \sqrt{x-7} + 2$

Endpoint: _____

3. $f(x) = 2\sqrt{x-1}$

Endpoint: _____

2. $f(x) = -\sqrt{x+3} - 4$

Endpoint: _____

4. $f(x) = \frac{1}{4}\sqrt{-x-8} - 5$

Endpoint: _____

Section 4: TransformationsDescribe ALL transformations from $y = \sqrt{x}$.

1. $f(x) = \sqrt{-x-5} - 3$

2. $f(x) = -2\sqrt{x-4}$

3. $f(x) = \frac{1}{3}\sqrt{2x-6} + 1$

4. $f(x) = 5\sqrt{x+2} - 8$

Section 5: Domain and Range

Find the domain and range of each function.

1. $f(x) = \sqrt{x-4} + 2$

D: _____ R: _____

2. $f(x) = -3\sqrt{x+1} - 5$

D: _____ R: _____

3. $f(x) = \frac{1}{2}\sqrt{x+7} + 3$

D: _____ R: _____

4. $f(x) = -\sqrt{x-2}$

D: _____ R: _____

Section 6: Finding the Inverse (Using the 4-Step Process)

Find the inverse of each function using the 4-step process.

1. $f(x) = -\sqrt{x+2} + 5$

A. $f^{-1}(x) = x^2 - 10x + 23$

B. $f^{-1}(x) = x^2 + 10x + 23$

C. $f^{-1}(x) = x^2 - 10x + 27$

D. $f^{-1}(x) = x^2 + 10x + 27$

3. $f(x) = -\frac{1}{2}\sqrt{x+3} - 2$

A. $f^{-1}(x) = 4x^2 - 16x + 13$

B. $f^{-1}(x) = 4x^2 + 16x + 13$

C. $f^{-1}(x) = 4x^2 + 16x + 19$

D. $f^{-1}(x) = 4x^2 - 16x + 19$

2. $f(x) = 3\sqrt{x-4} + 1$

A. $f^{-1}(x) = \frac{1}{9}x^2 - \frac{2}{9}x + \frac{37}{9}$

B. $f^{-1}(x) = \frac{1}{9}x^2 + \frac{2}{9}x + \frac{37}{9}$

C. $f^{-1}(x) = \frac{1}{9}x^2 - \frac{2}{9}x + \frac{35}{9}$

D. $f^{-1}(x) = 9x^2 - 2x + 37$

Section 8: Solving Square Root Equations

Solve each equation. Check for extraneous solutions!

1. $\sqrt{3x - 2} = 4$

4. $\sqrt{x + 3} = x + 1$

2. $\sqrt{x + 5} - 3 = 1$

5. $2(x + 1)^{1/2} - 6 = 0$

3. $\sqrt{2x - 1} + 2 = 7$

6. $\sqrt{4 - x} - 2 = x$